

CATAM Additional Projects

2.7 Soliton Solutions of the KdV Equation

2 Questions

Question 1

Firstly, we note the following:

$$\begin{aligned} f'(x) &= -\frac{2A}{\Delta} \operatorname{sech}^2\left(\frac{x-x_0}{\Delta}\right) \tanh\left(\frac{x-x_0}{\Delta}\right) \\ &= -\frac{2}{\Delta} f(x) \tanh\left(\frac{x-x_0}{\Delta}\right) \end{aligned}$$

$$\begin{aligned} f''(x) &= -\frac{2}{\Delta} f'(x) \tanh\left(\frac{x-x_0}{\Delta}\right) - \frac{2}{\Delta^2} f(x) \operatorname{sech}^2\left(\frac{x-x_0}{\Delta}\right) \\ &= \frac{f'(x)^2}{f(x)} - \frac{2}{\Delta^2 A} f(x)^2 \\ &= \frac{4}{\Delta^2} f(x) \tanh^2\left(\frac{x-x_0}{\Delta}\right) - \frac{2}{\Delta^2 A} f(x)^2 \end{aligned}$$

$$\begin{aligned} f'''(x) &= \frac{4}{\Delta^2} f'(x) \tanh^2\left(\frac{x-x_0}{\Delta}\right) \\ &\quad + \frac{8}{\Delta^3} f(x) \tanh\left(\frac{x-x_0}{\Delta}\right) \operatorname{sech}^2\left(\frac{x-x_0}{\Delta}\right) - \frac{4}{\Delta^2 A} f(x) f'(x) \\ &= \frac{4}{\Delta^2} f'(x) \tanh^2\left(\frac{x-x_0}{\Delta}\right) - \frac{4}{\Delta^2} f'(x) \operatorname{sech}^2\left(\frac{x-x_0}{\Delta}\right) - \frac{4}{\Delta^2 A} f(x) f'(x) \\ &= \frac{4}{\Delta^2} f'(x) \left[\tanh^2\left(\frac{x-x_0}{\Delta}\right) - \operatorname{sech}^2\left(\frac{x-x_0}{\Delta}\right) - \frac{1}{A} f(x) \right] \\ &= \frac{4}{\Delta^2} f'(x) \left[3 \tanh^2\left(\frac{x-x_0}{\Delta}\right) - 2 \right] \\ &= \frac{c}{\delta^2} f'(x) \left[3 \tanh^2\left(\frac{x-x_0}{\Delta}\right) - 2 \right]. \end{aligned}$$

We have:

- $u_t = -c f'(x - ct)$
- $uu_x = f(x - ct) f'(x - ct)$
- $\delta^2 u_{xxx} = c f'(x - ct) \left[3 \tanh^2\left(\frac{x-x_0-ct}{\Delta}\right) - 2 \right]$

$$\begin{aligned} \Rightarrow u_t + \delta^2 u_{xxx} &= c f'(x - ct) \left[3 \tanh^2\left(\frac{x-x_0-ct}{\Delta}\right) - 3 \right] \\ &= -3c f'(x - ct) \operatorname{sech}^2\left(\frac{x-x_0-ct}{\Delta}\right) \end{aligned}$$

$$\begin{aligned}
&= -f(x - ct)f'(x - ct) \\
&= -uu_{xx}
\end{aligned}$$

and hence,

$$u_t + uu_{xx} + \delta^2 u_{xxx} = 0 .$$

Now suppose we have a periodic solution $u(x, t)$ satisfying (3), then we will also have:

$$\begin{aligned}
u_x(x + 1, t) &= u_x(x, t) \\
u_{xx}(x + 1, t) &= u_{xx}(x, t) .
\end{aligned}$$

Hence,

$$\begin{aligned}
\frac{dM}{dt} &= \int_0^1 \frac{\partial u}{\partial t} dx \\
&= \int_0^1 -uu_x - \delta^2 u_{xxx} dx \\
&= \left[-\frac{1}{2}u^2 - \delta^2 u_{xx} \right]_0^1 \\
&= -\frac{1}{2}[u^2(1, t) - u^2(0, t)] - \delta^2[u_{xx}(1, t) - u_{xx}(0, t)] \\
&= 0 .
\end{aligned}$$

$$\begin{aligned}
\frac{dE}{dt} &= \int_0^1 u \frac{\partial u}{\partial t} dx \\
&= -\int_0^1 u(uu_x + \delta^2 u_{xxx}) dx \\
&= -\left[\frac{1}{2}u^3 + \delta^2 uu_{xx} \right]_0^1 + \int_0^1 \frac{1}{2}u^2 u_x + \delta^2 u_x u_{xx} dx \\
&= 0 + \left[\frac{1}{6}u^3 + \frac{\delta^2}{2}u_x^2 \right]_0^1 \\
&= 0 .
\end{aligned}$$

Thus both M and E are independent of time.

Question 2

The leap-frog scheme (5) is second order in both x & t , i.e. the truncation error is $u(hm, kn) - u_m^n = O(h^2) + O(k^2)$.

Let $\hat{x} = x/\delta$, $\hat{t} = t/\delta$ and $v(\hat{x}, \hat{t}) = u(x, t)$. Hence $\frac{\partial}{\partial x} = \frac{1}{\delta} \frac{\partial}{\partial \hat{x}}$ & $\frac{\partial}{\partial t} = \frac{1}{\delta} \frac{\partial}{\partial \hat{t}}$.

Now if $u(x, t)$ satisfies (1) we have

$$\begin{aligned} v_t(\hat{x}, \hat{t}) + v(\hat{x}, \hat{t})v_x(\hat{x}, \hat{t}) + \delta^2 v_{xxx}(\hat{x}, \hat{t}) &= 0 \\ \Rightarrow \frac{1}{\delta} v_{\hat{t}}(\hat{x}, \hat{t}) + \frac{1}{\delta} v(\hat{x}, \hat{t})v_{\hat{x}}(\hat{x}, \hat{t}) + \frac{\delta^2}{\delta^3} v_{\hat{x}\hat{x}\hat{x}}(\hat{x}, \hat{t}) &= 0 \\ \Rightarrow v_{\hat{t}}(\hat{x}, \hat{t}) + v(\hat{x}, \hat{t})v_{\hat{x}}(\hat{x}, \hat{t}) + v_{\hat{x}\hat{x}\hat{x}}(\hat{x}, \hat{t}) &= 0 \end{aligned}$$

Hence $v(\hat{x}, \hat{t})$ satisfies (1) with $\delta = 1$. Thus, the leap-frog scheme with v_m^n being the solution at $\hat{x} = \hat{h}m$ and $\hat{t} = \hat{k}n$ will be stable provided that

$$\begin{aligned} \hat{k} &\leq \frac{\hat{h}^3}{4 + \hat{h}^2 |v_{\max}|} \\ \Rightarrow \frac{k}{\delta} &\leq \frac{h/\delta^3}{4 + h/\delta^2 |v_{\max}|} \end{aligned}$$

Noting that $v_{\max} = u_{\max}$ we hence obtain the following stability condition for $\delta \neq 1$:

$$k \leq \frac{h^3}{4\delta^2 + h^2 |u_{\max}|}.$$

A program implementing (5) to solve (1) is listed on page 14. In order to calculate the first time step, the leap-frog scheme (5) cannot be used due to the dependence on u_m^{n-1} . Hence the following slightly altered scheme was used:

$$\begin{aligned} u_m^{n+1} &= u_m^n - \frac{k}{6h} (u_{m+1}^n + u_m^n + u_{m-1}^n)(u_{m+1}^n - u_{m-1}^n) \\ &\quad - \frac{k\delta^2}{2h^3} (u_{m+2}^n - 2u_{m+1}^n + 2u_{m-1}^n - u_{m-2}^n). \end{aligned}$$

In (5), u_t is discretised as $\frac{u_m^{n+1} - u_m^{n-1}}{2k}$ whereas as in the above scheme it is discretised as $\frac{u_m^{n+1} - u_m^n}{k}$. This eliminates the dependence on u_m^{n-1} whilst introducing a factor of $1/2$ to the other terms on the right-hand side.

To ensure numerical stability we use the stability condition for $\delta \neq 1$ derived above. Given the true soliton solution (2), we have $|u_{\max}| = A$ and hence in the case $A = 2$ & $\delta = 0.03$ this condition becomes:

$$k \leq \frac{h^3}{0.0036 + 2h^2}. \quad (\dagger)$$

Figure 2.1 plots the numerical and analytical solution of (1) with initial data (7) at $t = 0.5$ with $h = 0.0025$ & $k = 2.5 \times 10^{-6}$. Overall, the numerical solution closely

matches the analytical solution with only a small discrepancy visible in Figure 2.1.1 at the peak of the solutions. In Figure 2.1.2 however we see greater discrepancies at the tail where the numerical solution fluctuates somewhat randomly about analytical solution.

Figure 2.1 *Numerical & analytical solutions at $t = 0.5$*

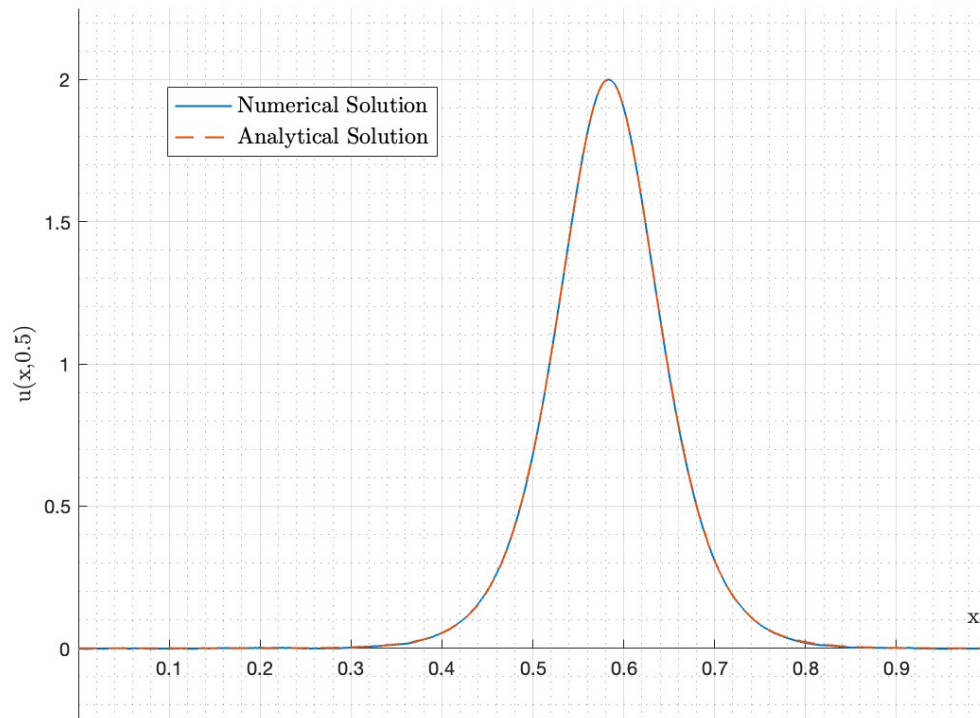


Figure 2.1.1 *A closer look at the peak of Figure 2.1*

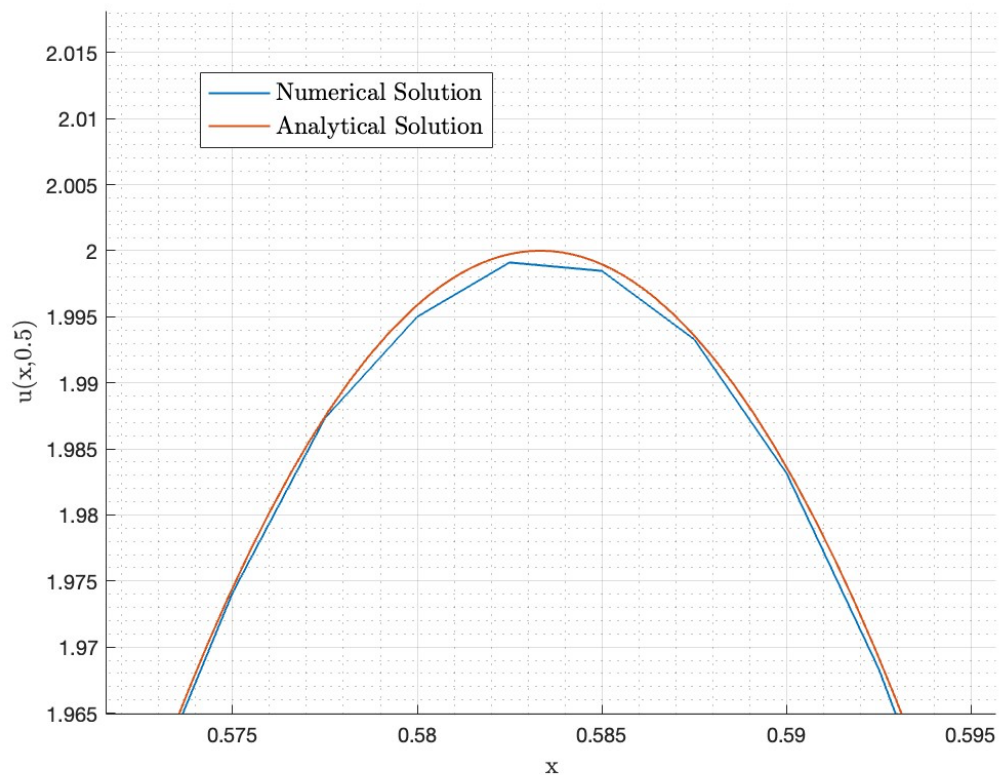
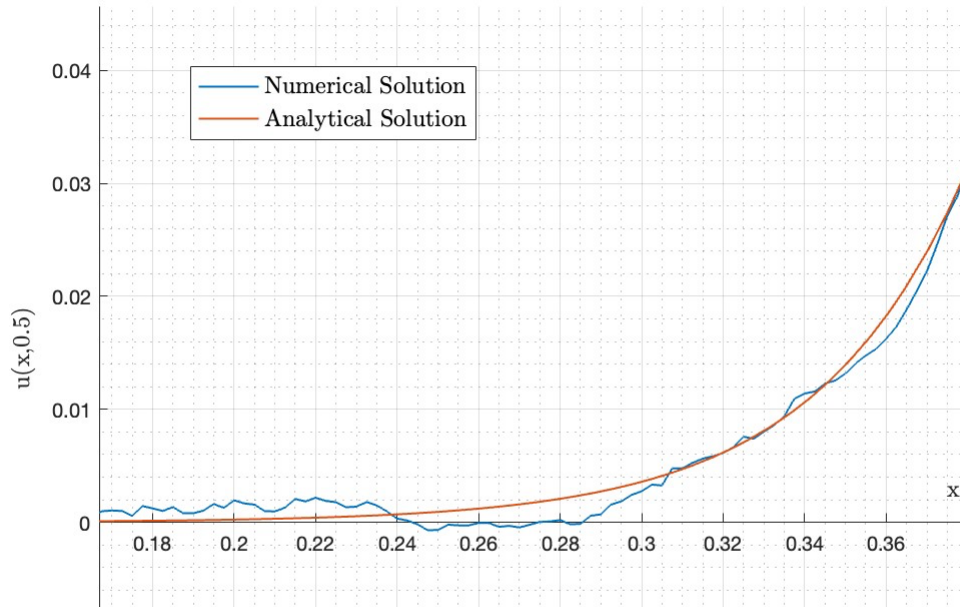
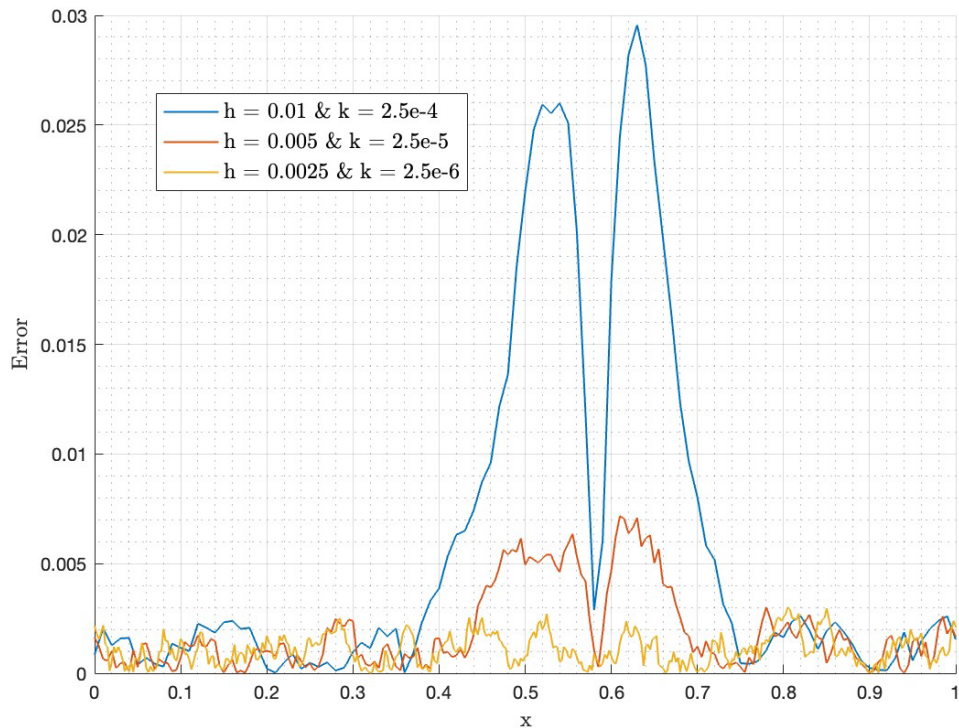


Figure 2.1.2 *A closer look at the tail of Figure 2.1*



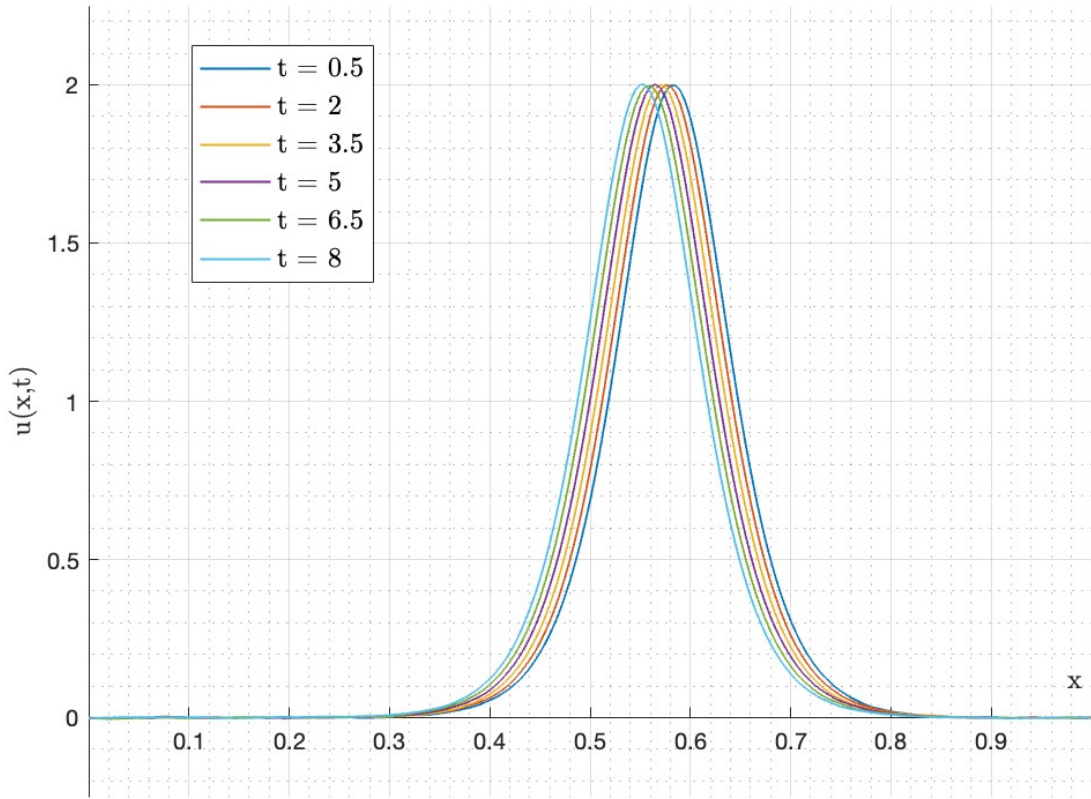
Illustrated in Figure 2.2 is the error $\epsilon = |u(hm, kn) - u_m^n|$ of the numerical solution for three pairs of decreasing h & k values. The value of h was specified and then an appropriate value of k was selected using $(\dagger)$. As we can see, the errors about the peak of the solution decrease as we decrease h & k , whilst the errors around tails of the solution remain a similar magnitude. Further reductions in the size of h (and hence k) resulted in little to no improvement in accuracy at the expense of greatly increased computational time. Hence the values $h = 0.0025$ & $k = 2.5 \times 10^{-6}$ were settled on for producing Figures 2.1, 2.1.1 & 2.1.2 as well as throughout the remainder of the project.

Figure 2.2 *Error of numerical solutions at $t = 0.5$ with decreasing h & k .*



The propagation speed of the analytical solution (2) with $A = 2$ is $c = 2/3$ and hence with the cyclic boundary conditions (3), we expect the solution to return to its starting position after a time $\Delta t = 1.5$. However, as we can see in Figure 2.3, this is not the case for the numerical solution. As t increases $t \mapsto t + 1.5$ the numerical solution doesn't quite reach the point where it started, and hence we deduce that propagation speed of the numerical solution is slightly less than that of the analytical solution.

Figure 2.3 Numerical solution at various times increasing by $\Delta t = 1.5$



Question 3

Now consider the two solitons:

$$f_1(x) = 2 \operatorname{sech}^2\left(\frac{x - 0.25}{\Delta_1}\right) \quad (A = 2, x_0 = 0.25)$$

$$f_2(x) = \operatorname{sech}^2\left(\frac{x - 0.75}{\Delta_2}\right) \quad (A = 1, x_0 = 0.75).$$

Figure 3.1 illustrates the evolution of the numerical solution with the initial data $u(x, 0) = f_1(x) + f_2(x)$ up to $t = 2.5$. Initially we can clearly identify the two solitons which begin travelling to the right. We notice that the larger amplitude soliton appears to be travelling at a greater speed and at around $t = 1$, catches up to the smaller amplitude soliton and by $t = 1.5$ has overtaken it. Hence the solution appears to be the superposition of two independently evolving solitons which move at different speeds. This, however, is not quite right. Indeed, this

cannot be the case since equation (1) has a nonlinear term uu_x and so solutions will not (in general) obey the superposition principle.

Illustrated in Figure 3.2 is the superposition of the two solutions with initial data $u(x, 0) = f_1(x)$ and $u(x, 0) = f_2(x)$ respectively. Whilst initially this superposition resembles the solution in Figure 3.1, by $t = 1.5$ they look different. This suggests that evolution of the two solitons in Figure 3.1 is not entirely independent and that the presence of both influences, for example, their speeds of propagation. Nevertheless, as t increases further, the solution in Figure 3.1 continues to consist of two identifiable solitons, one of amplitude 2 and the other of amplitude 1.

Figure 3.1 Numerical solution with initial data $u(x, 0) = f_1(x) + f_2(x)$ at various times

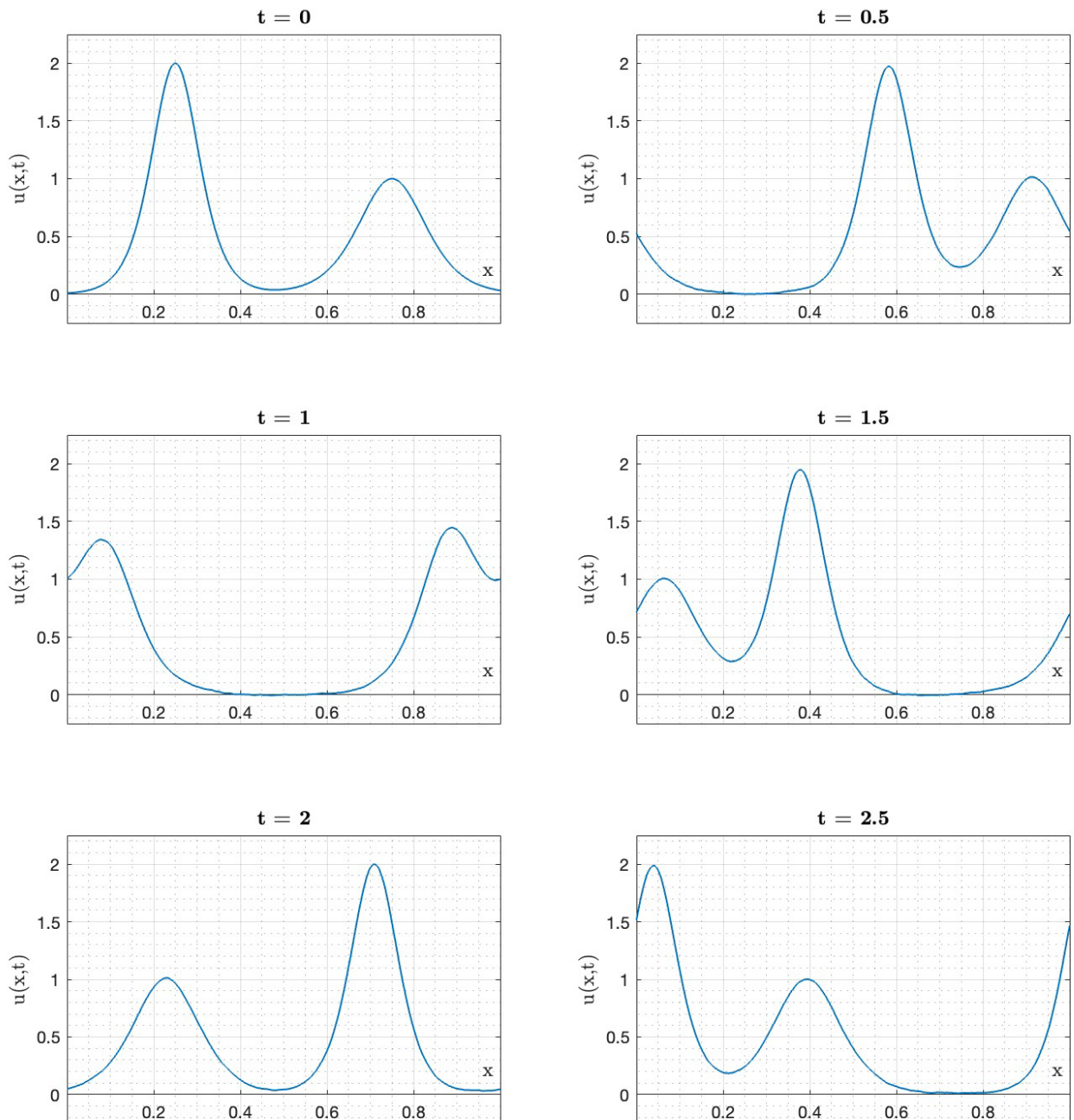
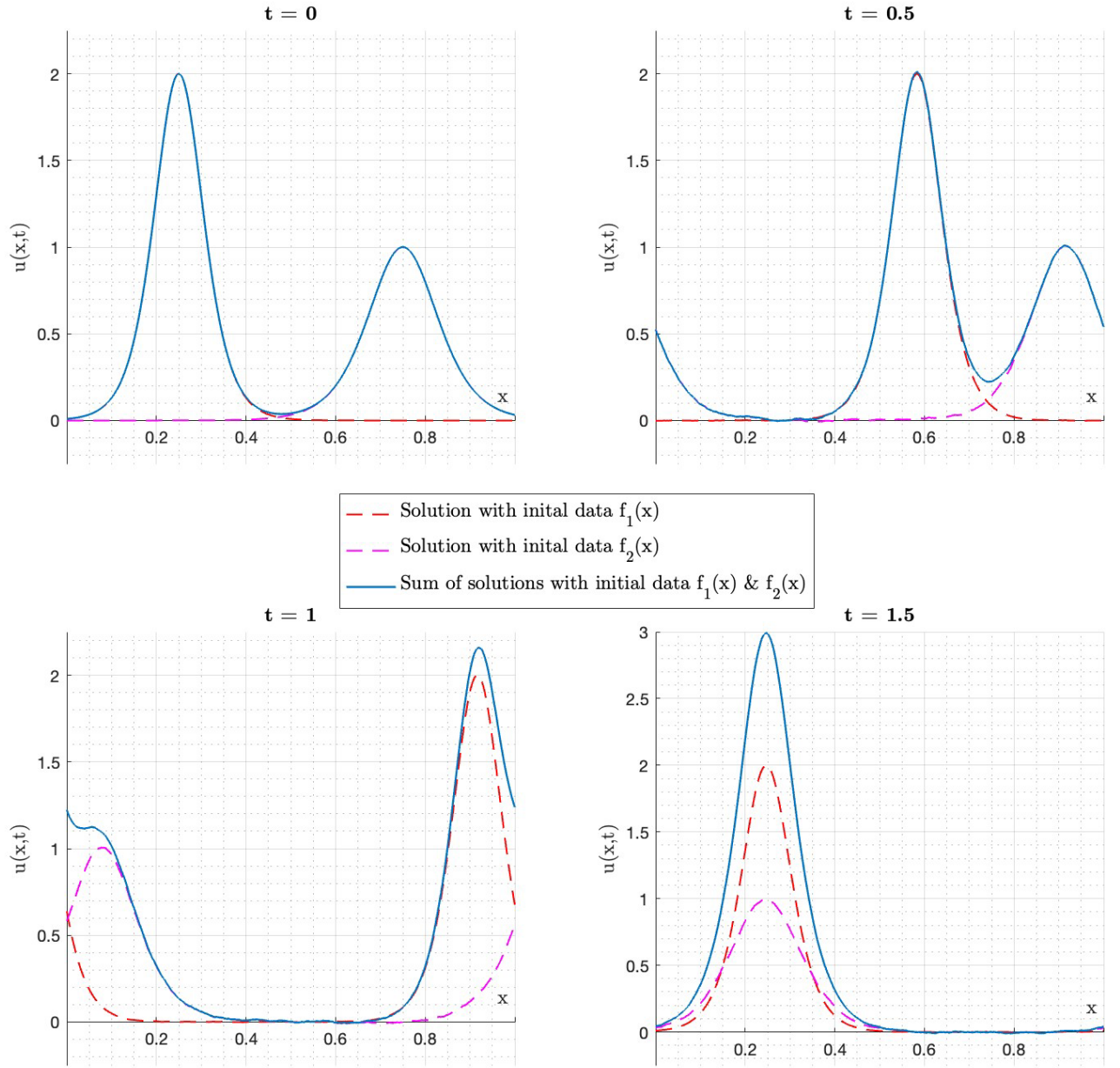


Figure 3.2 Numerical solutions with separate initial data $u(x, 0) = f_1(x)$ & $u(x, 0) = f_2(x)$



Using the numerical solutions found for the initial data $u(x, 0) = f_1(x) + f_2(x)$ we employ the trapezium rule to approximate quantities M & E defined in (4). Supposing $m = 0, 1, 2, \dots, \mathcal{M}$ then, at a given time $t = kn$, M is approximated by:

$$\begin{aligned}
 M &\approx h \sum_{m=1}^{\mathcal{M}} \frac{u_{m-1}^n + u_m^n}{2} \\
 &= h \sum_{m=0}^{\mathcal{M}-1} u_m^n
 \end{aligned}$$

and E is approximated by:

$$E \approx h \sum_{m=1}^{\mathcal{M}} \frac{\frac{1}{2}(u_{m-1}^n)^2 + \frac{1}{2}(u_m^n)^2}{2}$$

$$= \frac{h}{2} \sum_{m=0}^{\mathcal{M}-1} (u_m^n)^2.$$

In the second line of both equations, we have used the fact that $u(0, t) = u(1, t)$ and hence $u_0^n = u_{\mathcal{M}}^n$. As shown in question 1, M & E are independent of time and indeed, as illustrated in Table 3.1, our approximations using the numerical solution remain roughly constant in time.

Table 3.1 Numerical approximation to M & E

t	M	E
0	0.49975195	0.26543441
1	0.49732813	0.26418180
2	0.49972264	0.26543330
3	0.49982773	0.26543569
4	0.49959351	0.26542432
5	0.49729008	0.26414340

Question 4

In the case $\delta = 0$ the numerical solution with the initial data (8) begins by evolving smoothly up until about $t \approx 0.16$. At this point we see the outbreak of many smaller, sharper oscillations emanating from the midpoint $x = 0.5$, which suggest perhaps numerical instability or a breakdown of the analytical solution.

In the case $\delta = 0.03$, Figure 4.1 illustrates the evolution of the numerical solution with initial data (8). The x range has been increased to $0 < x < 2$ to make the form of the solutions clearer.

Initially, $\frac{\max|\delta^2 u_{xxx}|}{\max|uu_x|} = \frac{0.03^2(2\pi)^3}{2\pi} \approx 0.0355$ and so it is the uu_x term in (1) that dominates. As we saw for $\delta = 0$, in the absence of a diffusive term, the effect of the nonlinear term uu_x is to steepen the wavefronts. Indeed. In Figure 4.1, this is initially what we see, with the wavefronts at $x = 0.5$ & $x = 1.5$ steepening.

As these wavefronts steepen however, the diffusive $\delta^2 u_{xxx}$ term becomes more significant and results in the emergence of narrower, soliton-like, wave peaks at around $t = 0.15$. By $t = 0.5$ three clear solitons have emerged with the largest soliton propagating to the right and the two smaller solitons propagating to the left. Similarly to what we saw in question 3, these solitons move and interact with one another whilst retaining their single wave-peak like shape after each interaction.

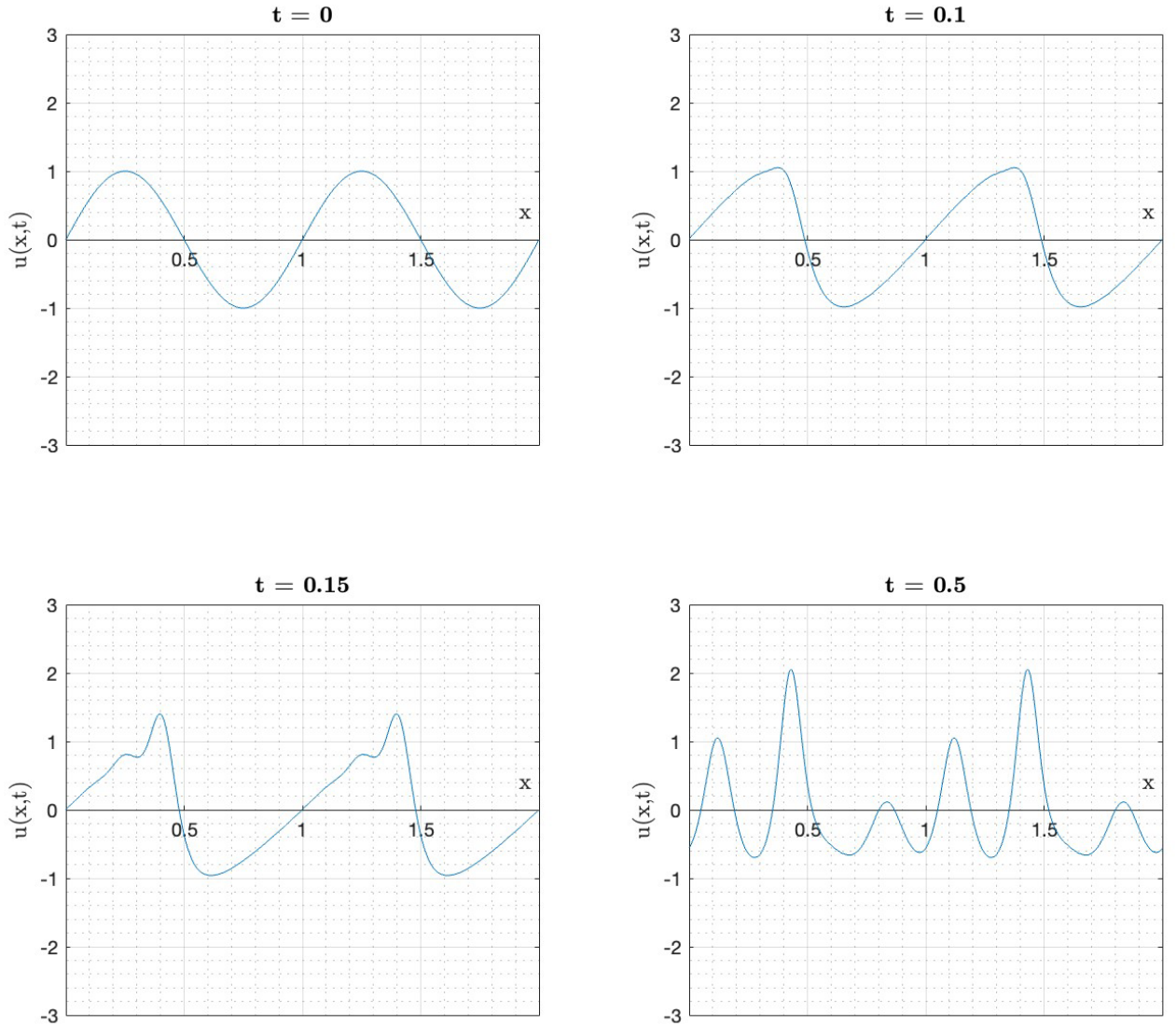
This change in character of the solution occurs once the steepening effect of the uu_x term has caused the $\delta^2 u_{xxx}$ term to become significant. Hence, to estimate the time at which the solution changes character we wish to determine the time scale T over which the steepening occurs.

Given the initial data (8), the characteristic length scale over which the solution changes is $L = \frac{1}{2\pi}$ and the scale of the solution itself is 1. Approximating the initial equation by $u_t + uu_x \approx 0$ and substituting in these length and time scales we have:

$$\frac{1}{T} \sim \frac{1}{L},$$

and hence $T \approx \frac{1}{2\pi} \approx 0.16$.

Figure 4.1 Numerical solution with initial data (8) and $\delta = 0.03$ at various times



Figures 4.2 & 4.3 illustrate the numerical solution in the cases $\delta = 0.02$ & $\delta = 0.04$ respectively. The evolution of the solutions is essentially the same as in Figure 4.1, with the change in δ resulting in a change in the number of solitons that emerge. Indeed, for a larger diffusivity ($\delta = 0.04$) fewer solitons emerge which are broader, whilst for a smaller diffusivity ($\delta = 0.02$) more solitons emerge which are narrower and sharper. As seen before, the solitons always emerge in a line of increasing amplitude with the larger amplitude solitons propagating at greater speeds, overtaking the smaller amplitude solitons.

Interestingly, we notice in Figures 4.1 – 4.3 the emergence of solitons at $t \approx 0.15$, which is slightly earlier than our estimate $T \approx \frac{1}{2\pi}$. This is because, as δ increases, the diffusive term will become more significant sooner and hence we expect to observe the emergence of solitons at slightly earlier times. Indeed, our estimate $T \approx \frac{1}{2\pi} \approx 0.16$ agrees most closely with what was observed in the case $\delta = 0$ since, in constructing this estimate, we used (1) with $\delta = 0$.

Figure 4.2 Numerical solution with initial data (8) and $\delta = 0.02$ at various times

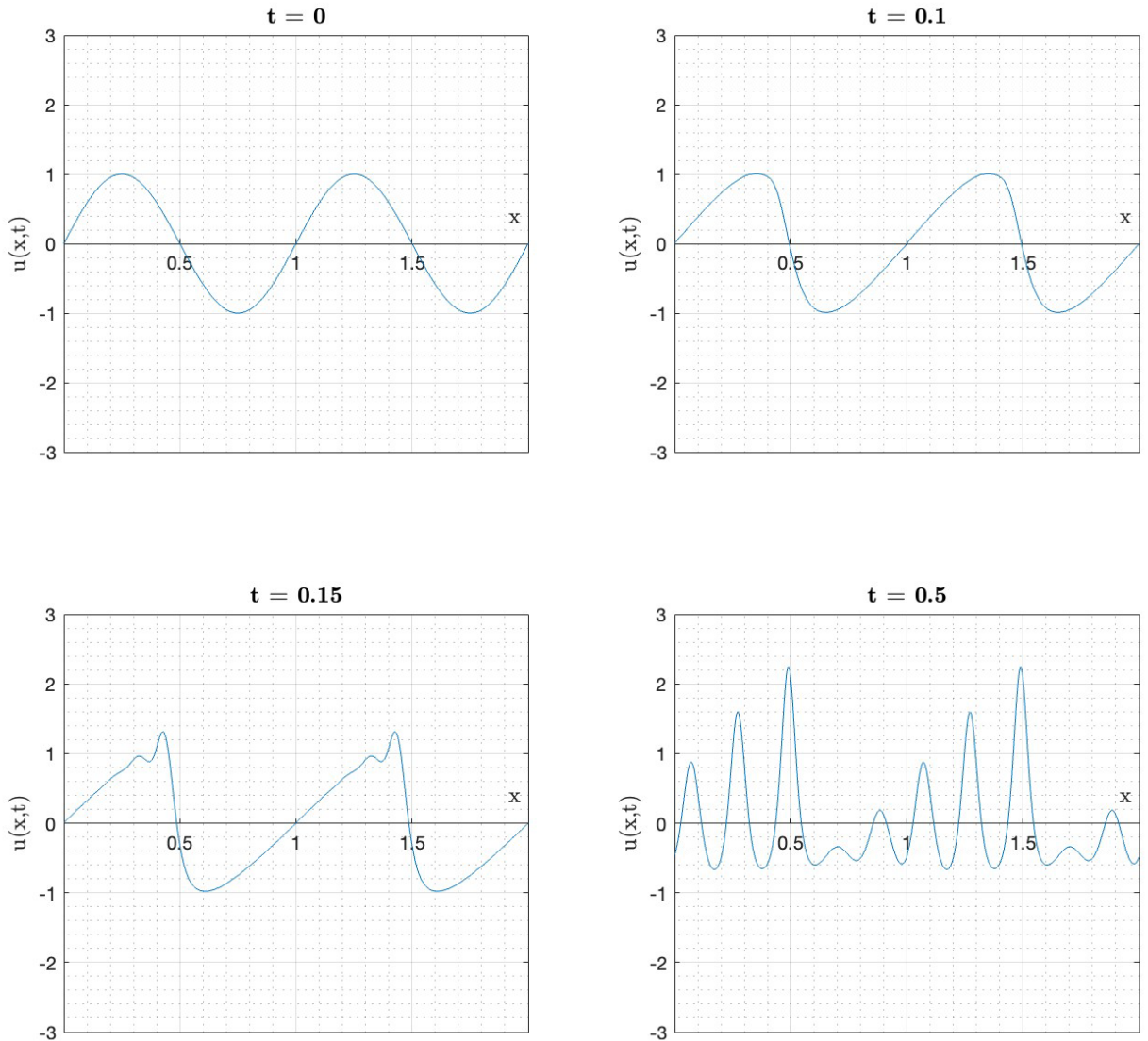
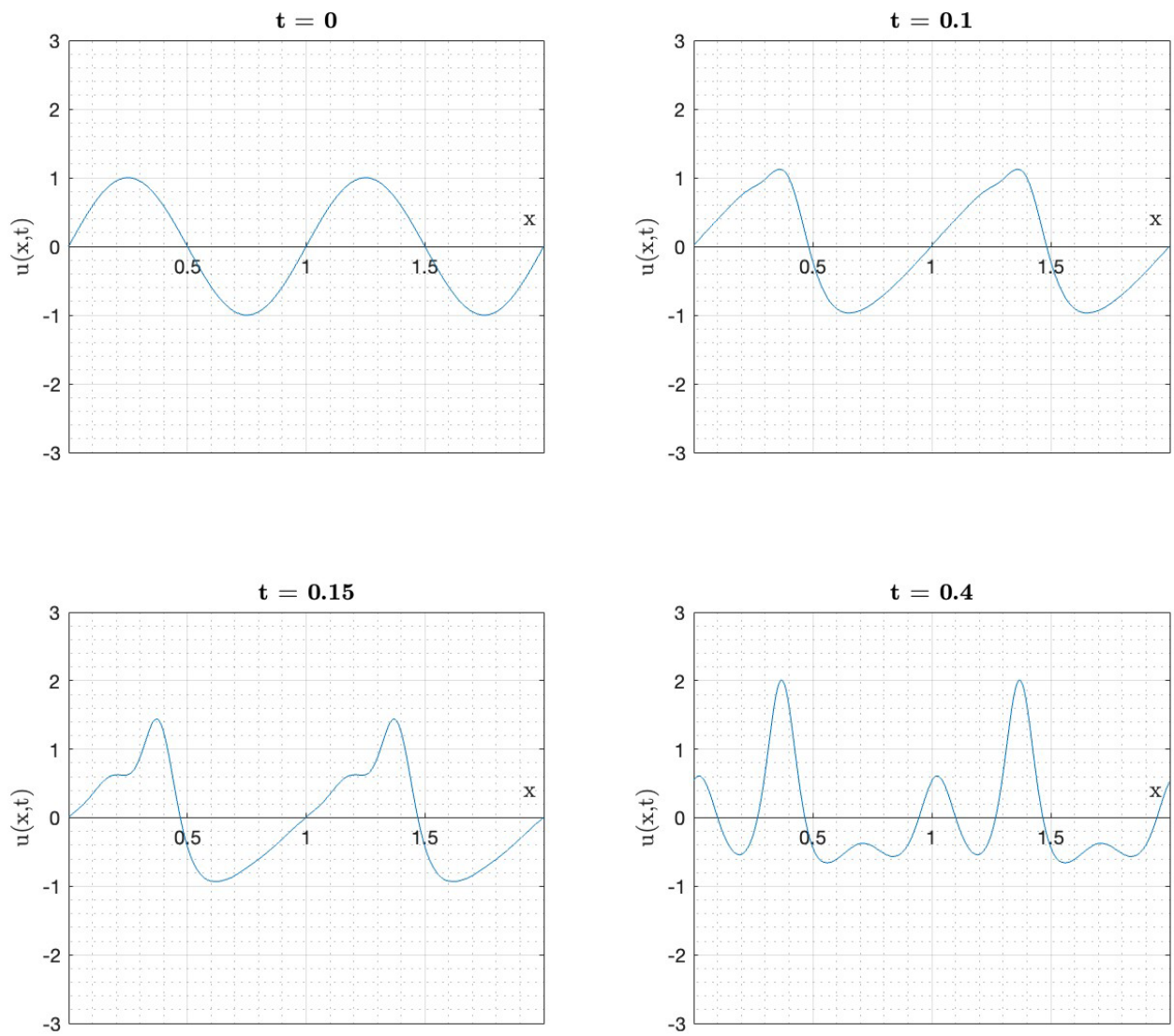


Figure 4.3 *Numerical solution with initial data (8) and $\delta = 0.04$ at various times*



Programs

All code written in MATLAB version R2024b

Question 2

Program implementing (5) to solve (1)

```
function u = KdVQuestion2(delta,h,k,t,f)
% Solves (1) by implementing (5)
% INPUTS:
% delta
% h - size of x step
% k - size of t step
% t - value of t up to which we are evaluating u
% f - function corresponding to u(x,0)
% OUTPUTS:
% u - array of values corresponding to u(mh,nk)

M = 1/h;
N = t/k;
% Establishing initial conditions
u = zeros(M+1,N+1);
for i = 1:M+1
    u(i,1) = f((i-1)*h);
end
% First time step with alternative method
u(1,2) = u(1,1) - (k/(6*h))*(u(1+1,1)+u(1,1)+u(end+1-1,1))*(u(1+1,1)-
u(end+1-1,1))-(k*(delta^2)/(2*(h^3)))*(u(1+2,1)-2*u(1+1,1)+2*u(end+1-1,1)-
u(end+1-2,1));
u(2,2) = u(2,1) - (k/(6*h))*(u(2+1,1)+u(2,1)+u(2-1,1))*(u(2+1,1)-u(2-
1,1))-(k*(delta^2)/(2*(h^3)))*(u(2+2,1)-2*u(2+1,1)+2*u(2-1,1)-u(end+2-
2,1));
for i = 3:M-1
    u(i,2) = u(i,1) - (k/(6*h))*(u(i+1,1)+u(i,1)+u(i-1,1))*(u(i+1,1)-u(i-
1,1))-(k*(delta^2)/(2*(h^3)))*(u(i+2,1)-2*u(i+1,1)+2*u(i-1,1)-u(i-2,1));
end
u(M,2) = u(M,1) - (k/(6*h))*(u(M+1,1)+u(M,1)+u(M-1,1))*(u(M+1,1)-u(M-
1,1))-(k*(delta^2)/(2*(h^3)))*(u(1,1)-2*u(M+1,1)+2*u(M-1,1)-u(M-2,1));
u(M+1,2) = u(M+1,1) - (k/(6*h))*(u(1,1)+u(M+1,1)+u(M+1-1,1))*(u(1,1)-
u(M+1-1,1))-(k*(delta^2)/(2*(h^3)))*(u(2,1)-2*u(1,1)+2*u(M+1-1,1)-u(M+1-
2,1));
% Remaining time steps using (5)
for j = 2:N
    u(1,j+1) = u(1,j-1) - (k/(3*h))*(u(1+1,j)+u(1,j)+u(end+1-
1,j))*(u(1+1,j)-u(end+1-1,j))-(k*(delta^2)/(h^3))*(u(1+2,j)-
2*u(1+1,j)+2*u(end+1-1,j)-u(end+1-2,j));
    u(2,j+1) = u(2,j-1) - (k/(3*h))*(u(2+1,j)+u(2,j)+u(2-1,j))*(u(2+1,j)-
u(2-1,j))-(k*(delta^2)/(h^3))*(u(2+2,j)-2*u(2+1,j)+2*u(2-1,j)-u(end,j));
    for i = 3:M-1
        u(i,j+1) = u(i,j-1) - (k/(3*h))*(u(i+1,j)+u(i,j)+u(i-1,j))*(u(i+1,j)-
u(i-1,j))-(k*(delta^2)/(h^3))*(u(i+2,j)-2*u(i+1,j)+2*u(i-1,j)-u(i-2,j));
    end
    u(M,j+1) = u(M,j-1) - (k/(3*h))*(u(M+1,j)+u(M,j)+u(M-1,j))*(u(M+1,j)-
u(M-1,j))-(k*(delta^2)/(h^3))*(u(1,j)-2*u(M+1,j)+2*u(M-1,j)-u(M-2,j));
    u(M+1,j+1) = u(M+1,j-1) - (k/(3*h))*(u(1,j)+u(M+1,j)+u(M+1-
1,j))*(u(1,j)-u(M+1-1,j))-(k*(delta^2)/(h^3))*(u(2,j)-2*u(1,j)+2*u(M+1-
1,j)-u(M+1-2,j));
end
```


Program used to produce Figures 2.1, 2.1.1 & 2.1.2

```
D = sqrt((12*(0.03^2))/2);
t = 0.5;
k = 0.0000025;
h = 0.0025;
u = KdVQuestion2(0.03, h, k, t, @(x) 2*sech((x-0.25)/D)^2);

hold on
plot(0:h:1,u(:,end),'LineWidth', 1)
fplot(@(x) 2*(sech((x-(2/3)*t-0.25)/D))^2, [0 1],'LineWidth', 1)
ylim([-0.25 2.25])
ax = gca;
ax.XAxisLocation = 'origin';
grid(ax,"on")
grid(ax,'minor')
xlabel('x', 'FontName', 'CMU Serif', 'FontSize', 12)
ylabel('u(x,0.5)', 'FontName', 'CMU Serif', 'FontSize', 12)
legend('Numerical Solution', 'Analytical Solution', 'FontName', 'CMU
Serif', 'FontSize', 12)
```

Program used to produce Figure 2.2

```
D = sqrt((12*(0.03^2))/2);
t = 0.5;
k = [0.00025;2.5e-5;2.5e-6];
h = [0.01;0.005;0.0025];
hold on
for j = 1:3
    u = KdVQuestion2(0.03, h(j), k(j), t, @(x) 2*sech((x-0.25)/D)^2);
    f = @(x) 2*(sech((x-(2/3)*t-0.25)/D))^2;
    F = zeros(round(1/h(j)) + 1,1);
    for i = 1:round(1/h(j)) + 1
        F(i) = f((i-1)*h(j));
    end
    plot(0:h(j):1,abs(u(:,end)-F),'LineWidth', 1)
end
ax = gca;
ax.XAxisLocation = 'origin';
grid(ax,"on")
grid(ax,'minor')
xlabel('x', 'FontName', 'CMU Serif', 'FontSize', 12)
ylabel('Error', 'FontName', 'CMU Serif', 'FontSize', 12)
legend('h = 0.01 & k = 2.5e-4', 'h = 0.005 & k = 2.5e-5', 'h = 0.0025 & k =
2.5e-6', 'FontName', 'CMU Serif', 'FontSize', 12)
```

Program used to produce Figure 2.3

```
D = sqrt((12*(0.03^2))/2);
t = 8;
k = 0.0000025;
h = 0.005;
u = KdVQuestion2(0.03, h, k, t, @(x) 2*sech((x-0.25)/D)^2);

hold on
plot(0:h:1,u(:,0.5/k),'LineWidth', 0.9)
plot(0:h:1,u(:,2/k),'LineWidth', 0.9)
plot(0:h:1,u(:,3.5/k),'LineWidth', 0.9)
plot(0:h:1,u(:,5/k),'LineWidth', 0.9)
plot(0:h:1,u(:,6.5/k),'LineWidth', 0.9)
```

```

plot(0:h:1,u(:,8/k),'LineWidth', 0.9)
ylim([-0.25 2.25])
ax = gca;
ax.XAxisLocation = 'origin';
grid(ax,"on")
grid(ax,'minor')
xlabel('x', 'FontName', 'CMU Serif', 'FontSize', 12)
ylabel('u(x,t)', 'FontName', 'CMU Serif', 'FontSize', 12)
legend('t = 0.5', 't = 2', 't = 3.5', 't = 5', 't = 6.5', 't = 8',
'FontName', 'CMU Serif', 'FontSize', 12)

```

Question 3

Program used to produce Figure 3.1

```

D1 = sqrt((12*(0.03^2))/2);
D2 = sqrt((12*(0.03^2)));
t = 2.5;
k = 0.0000025;
h = 0.0025;
u1 = KdVQuestion2(0.03, h, k, t, @(x) 2*sech((x-0.25)/D1)^2 + sech((x-
0.75)/D2)^2);

subplot(3,2,1)
plot(0:h:1,u1(:,1),'LineWidth', 1)
ylim([-0.25 2.25])
ax = gca;
ax.XAxisLocation = 'origin';
grid(ax,"on")
grid(ax,'minor')
xlabel('x', 'FontName', 'CMU Serif', 'FontSize', 12)
ylabel('u(x,t)', 'FontName', 'CMU Serif', 'FontSize', 12)
title('t = 0', 'FontName', 'CMU Serif', 'FontWeight','bold', 'FontSize',
12)

subplot(3,2,2)
plot(0:h:1,u1(:,round(0.5/k)+1),'LineWidth', 1)
ylim([-0.25 2.25])
ax = gca;
ax.XAxisLocation = 'origin';
grid(ax,"on")
grid(ax,'minor')
xlabel('x', 'FontName', 'CMU Serif', 'FontSize', 12)
ylabel('u(x,t)', 'FontName', 'CMU Serif', 'FontSize', 12)
title('t = 0.5', 'FontName', 'CMU Serif', 'FontWeight','bold', 'FontSize',
12)

subplot(3,2,3)
plot(0:h:1,u1(:,round(1/k)+1),'LineWidth', 1)
ylim([-0.25 2.25])
ax = gca;
ax.XAxisLocation = 'origin';
grid(ax,"on")
grid(ax,'minor')
xlabel('x', 'FontName', 'CMU Serif', 'FontSize', 12)
ylabel('u(x,t)', 'FontName', 'CMU Serif', 'FontSize', 12)
title('t = 1', 'FontName', 'CMU Serif', 'FontWeight','bold', 'FontSize',
12)

subplot(3,2,4)

```



```

plot(0:h:1,u1(:,round(1.5/k)+1),'LineWidth', 1)
ylim([-0.25 2.25])
ax = gca;
ax.XAxisLocation = 'origin';
grid(ax,"on")
grid(ax,'minor')
xlabel('x', 'FontName', 'CMU Serif', 'FontSize', 12)
ylabel('u(x,t)', 'FontName', 'CMU Serif', 'FontSize', 12)
title('t = 1.5', 'FontName', 'CMU Serif', 'FontWeight','bold', 'FontSize',
12)

subplot(3,2,5)
plot(0:h:1,u1(:,round(2/k)+1),'LineWidth', 1)
ylim([-0.25 2.25])
ax = gca;
ax.XAxisLocation = 'origin';
grid(ax,"on")
grid(ax,'minor')
xlabel('x', 'FontName', 'CMU Serif', 'FontSize', 12)
ylabel('u(x,t)', 'FontName', 'CMU Serif', 'FontSize', 12)
title('t = 2', 'FontName', 'CMU Serif', 'FontWeight','bold', 'FontSize',
12)

subplot(3,2,6)
plot(0:h:1,u1(:,round(2.5/k)+1),'LineWidth', 1)
ylim([-0.25 2.25])
ax = gca;
ax.XAxisLocation = 'origin';
grid(ax,"on")
grid(ax,'minor')
xlabel('x', 'FontName', 'CMU Serif', 'FontSize', 12)
ylabel('u(x,t)', 'FontName', 'CMU Serif', 'FontSize', 12)
title('t = 2.5', 'FontName', 'CMU Serif', 'FontWeight','bold', 'FontSize',
12)

```

Program used to produce Figure 3.2

```

D1 = sqrt((12*(0.03^2))/2);
D2 = sqrt((12*(0.03^2)));
t = 1.5;
k = 0.0000025;
h = 0.0025;
u2 = KdVQuestion2(0.03, h, k, t, @(x) 2*sech((x-0.25)/D1)^2);
u3 = KdVQuestion2(0.03, h, k, t, @(x) sech((x-0.75)/D2)^2);

subplot(2,2,1)
hold on
plot(0:h:1,u2(:,1),'LineWidth', 1, 'LineStyle', '--', 'Color', 'r')
plot(0:h:1,u3(:,1),'LineWidth', 1, 'LineStyle', '--', 'Color', 'm')
plot(0:h:1,u2(:,1)+u3(:,1),'LineWidth', 1.2, 'Color', '#0072BD')
ylim([-0.25 2.25])
ax = gca;
ax.XAxisLocation = 'origin';
grid(ax,"on")
grid(ax,'minor')
xlabel('x', 'FontName', 'CMU Serif', 'FontSize', 12)
ylabel('u(x,t)', 'FontName', 'CMU Serif', 'FontSize', 12)
title('t = 0', 'FontName', 'CMU Serif', 'FontWeight','bold', 'FontSize',
12)

```

```

legend( "Solution with initial data f_1(x)", "Solution with initial data
f_2(x)", 'Sum of solutions with initial data f_1(x) & f_2(x)', 'FontName',
'CMU Serif', 'FontSize', 12)

subplot(2,2,2)
hold on
plot(0:h:1,u2(:,round(0.5/k)+1),'LineWidth', 1, 'LineStyle', '--',
'Color', 'r')
plot(0:h:1,u3(:,round(0.5/k)+1),'LineWidth', 1, 'LineStyle', '--',
'Color', 'm')
plot(0:h:1,u2(:,round(0.5/k)+1)+u3(:,round(0.5/k)+1),'LineWidth', 1.2,
'Color', '#0072BD')
ylim([-0.25 2.25])
ax = gca;
ax.XAxisLocation = 'origin';
grid(ax,"on")
grid(ax,'minor')
xlabel('x', 'FontName', 'CMU Serif', 'FontSize', 12)
ylabel('u(x,t)', 'FontName', 'CMU Serif', 'FontSize', 12)
title('t = 0.5', 'FontName', 'CMU Serif', 'FontWeight','bold', 'FontSize',
12)

subplot(2,2,3)
hold on
plot(0:h:1,u2(:,round(1/k)+1),'LineWidth', 1, 'LineStyle', '--', 'Color',
'r')
plot(0:h:1,u3(:,round(1/k)+1),'LineWidth', 1, 'LineStyle', '--',
'Color', 'm')
plot(0:h:1,u2(:,round(1/k)+1)+u3(:,round(1/k)+1),'LineWidth', 1.2,
'Color', '#0072BD')
ylim([-0.25 2.25])
ax = gca;
ax.XAxisLocation = 'origin';
grid(ax,"on")
grid(ax,'minor')
xlabel('x', 'FontName', 'CMU Serif', 'FontSize', 12)
ylabel('u(x,t)', 'FontName', 'CMU Serif', 'FontSize', 12)
title('t = 1', 'FontName', 'CMU Serif', 'FontWeight','bold', 'FontSize',
12)

subplot(2,2,4)
hold on
plot(0:h:1,u2(:,round(1.5/k)+1),'LineWidth', 1, 'LineStyle', '--',
'Color', 'r')
plot(0:h:1,u3(:,round(1.5/k)+1),'LineWidth', 1, 'LineStyle', '--',
'Color', 'm')
plot(0:h:1,u2(:,round(1.5/k)+1)+u3(:,round(1.5/k)+1),'LineWidth', 1.2,
'Color', '#0072BD')
ylim([-0.25 3])
ax = gca;
ax.XAxisLocation = 'origin';
grid(ax,"on")
grid(ax,'minor')
xlabel('x', 'FontName', 'CMU Serif', 'FontSize', 12)
ylabel('u(x,t)', 'FontName', 'CMU Serif', 'FontSize', 12)
title('t = 1.5', 'FontName', 'CMU Serif', 'FontWeight','bold', 'FontSize',
12)

```

Program used to produce Table 3.1

```
D1 = sqrt((12*(0.03^2))/2);
D2 = sqrt((12*(0.03^2)));
k = 0.0000025;
h = 0.0025;
t = [0;1;2;3;4;5];
u = KdVQuestion2(0.03, h, k, t(end), @(x) 2*sech((x-0.25)/D1)^2 + sech((x-0.75)/D2)^2);

M = zeros(length(t),1);
for i = 1:length(t)
    u_t = u(1:end-1, round(t(i)/k)+1);
    M(i) = h*sum(u_t);
end

E = zeros(length(t),1);
for i = 1:length(t)
    u_t = u(1:end-1, round(t(i)/k)+1);
    E(i) = (h/2)*sum(u_t.^2);
end

table(t, M, E)
```

Question 4

Program used to produce Figure 4.1

```
k = 0.0000025;
h = 0.0025;
t = [0;0.1;0.15;0.5;1.15;2.2];
for i = 1:length(t)
    u = KdVQuestion2(0.03, h, k, t(end), @(x) sin(2*pi*x));
    subplot(3,2,i)
    u2 = [u(:,round(t(i)/k)+1); u(2:end,round(t(i)/k)+1)];
    plot(0:h:2, u2)
    ylim([-3 3])
    ax = gca;
    ax.XAxisLocation = 'origin';
    grid(ax,"on")
    grid(ax,'minor')
    xlabel('x', 'FontName', 'CMU Serif', 'FontSize', 12)
    ylabel('u(x,t)', 'FontName', 'CMU Serif', 'FontSize', 12)
    str = "t = " + num2str(t(i));
    title(str, 'FontName', 'CMU Serif', 'FontSize', 12)
end
```

Program used to produce Figure 4.2

```
k = 0.0000025;
h = 0.0025;
t = [0;0.1;0.15;0.5];
for i = 1:length(t)
    u = KdVQuestion2(0.02, h, k, t(end), @(x) sin(2*pi*x));
    subplot(2,2,i)
    u2 = [u(:,round(t(i)/k)+1); u(2:end,round(t(i)/k)+1)];
    plot(0:h:2, u2)
    ylim([-3 3])
    ax = gca;
    ax.XAxisLocation = 'origin';
    grid(ax,"on")
```

```

grid(ax,'minor')
xlabel('x', 'FontName', 'CMU Serif', 'FontSize', 12)
ylabel('u(x,t)', 'FontName', 'CMU Serif', 'FontSize', 12)
str = "t = " + num2str(t(i));
title(str, 'FontName', 'CMU Serif', 'FontSize', 12)
end

```

Program used to produce Figure 4.3

```

k = 0.0000025;
h = 0.0025;
t = [0;0.1;0.15;0.4];
for i = 1:length(t)
    u = KdVQuestion2(0.04, h, k, t(end), @(x) sin(2*pi*x));
    subplot(2,2,i)
    u2 = [u(:,round(t(i)/k)+1); u(2:end,round(t(i)/k)+1)];
    plot(0:h:2, u2)
    ylim([-3 3])
    ax = gca;
    ax.XAxisLocation = 'origin';
    grid(ax,"on")
    grid(ax,'minor')
    xlabel('x', 'FontName', 'CMU Serif', 'FontSize', 12)
    ylabel('u(x,t)', 'FontName', 'CMU Serif', 'FontSize', 12)
    str = "t = " + num2str(t(i));
    title(str, 'FontName', 'CMU Serif', 'FontSize', 12)
end

```